Renaud Dessalles

Engineer, PhD in Applied Mathematics



With a background in engineering and research, I have multidisciplinary expertise across many scientific fields. I specialize in probabilistic, statistical, and deterministic methods, as well as in computer simulations applied in particular to various areas of life sciences, collaborating with biologists, agronomists, and physicists in an international context.

Profile

1989 (35 ans)

Kremelin-Bicêtre (94)

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https://dessalles.github.io/

renaud-dessalles-910960171

github.com/dessalles/

Languages

- French — native
- Anglais
- Espagnol
- Japonais

Skills

- Expert in Python, NumPy & Matplotlib
- Probabilistic modeling
- Deterministic modeling
- Mathematical analysis
- Simulations
- Data processing
- Optimal control
- Machine learning
- Geographic Information Systems (GIS)
- Interface with life sciences

Strengths

- Interdisciplinary approach
- Scientific communication & mediation
- Adaptability
- *Bridging theory and practice*

Interests

- Triathlon
- Short story writing
- World history

Professional Experience

2019–2025 – Head of the Modeling Division, Greenshield (Paris)
Coordinated the modeling team (6 members).
Supervised junior profiles (interns and apprentices).
Modeled crop pests and developed spray maps.
Designed an optimization platform to reduce pesticide use.

2017-2019 – Postdoctoral Researcher in Applied Mathematics and Physics for Immunology (University of California, Los Angeles – USA).
Developed models of T-cell diversity.
Created population models applied to 20th-century China.
Built reaction–diffusion models: development of bat teeth.

2017-2019 – Workshop Instructor for Students, PhD Candidates, and Researchers (University of California, Los Angeles – USA)
Introdution to Python
Machine Learning
Python Hackathon

2013-2016 – PhD in Applied Mathematics (INRA and INRIA – Paris)
Stochastic modeling of protein production in bacteria.

2015-2016 – TA in Mathematics and Computer Science (IUT Sceaux)

2013 – 4-month internship at LCQB lab (UPMC), Theoretical Microbiology.

2012 – 4-month internship at SMILE (UPMC), Ecology and Evolution.

2011 – 4-month internship at U1001 (INSERM), Systems Biology.

La thèse, la liste d'articles et conférences scientifiques disponibles sur dessalles.github.io/

Education

2017 – PhD : Applied Mathematics for Systems Biology (INRA and INRIA – Paris).

2013 – Master 2 : Mathematics for Life Sciences (Université ParisSud)

2012 – Engineering Degree : Systems Modeling (ENSTA ParisTech)

2012 – Master 2 : Numerical Analysis of Partial Differential Equations (UPMC)

Personal Projects

2025 – EventualMe : Android application for self-evaluating prediction accuracy (Dart, Flutter).

2015-2019 – RFigure : Python software for saving and editing scientific figures (Python, NumPy, Matplotlib).

2012-2015 – AthenaWriter : Short story text editor (Python, PyQT).

Source code available on my Github .